

# eProject Guide
# Building Apps for Android Devices

# eProject Guide - Building Apps for Android Devices

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# Preface

The eProject guide provides all the necessary information to students in developing a solution for the work process. Important functionalities required in the application are explained. Students have to understand the workflow and design the project accordingly.

The project specification is organized in the following order:

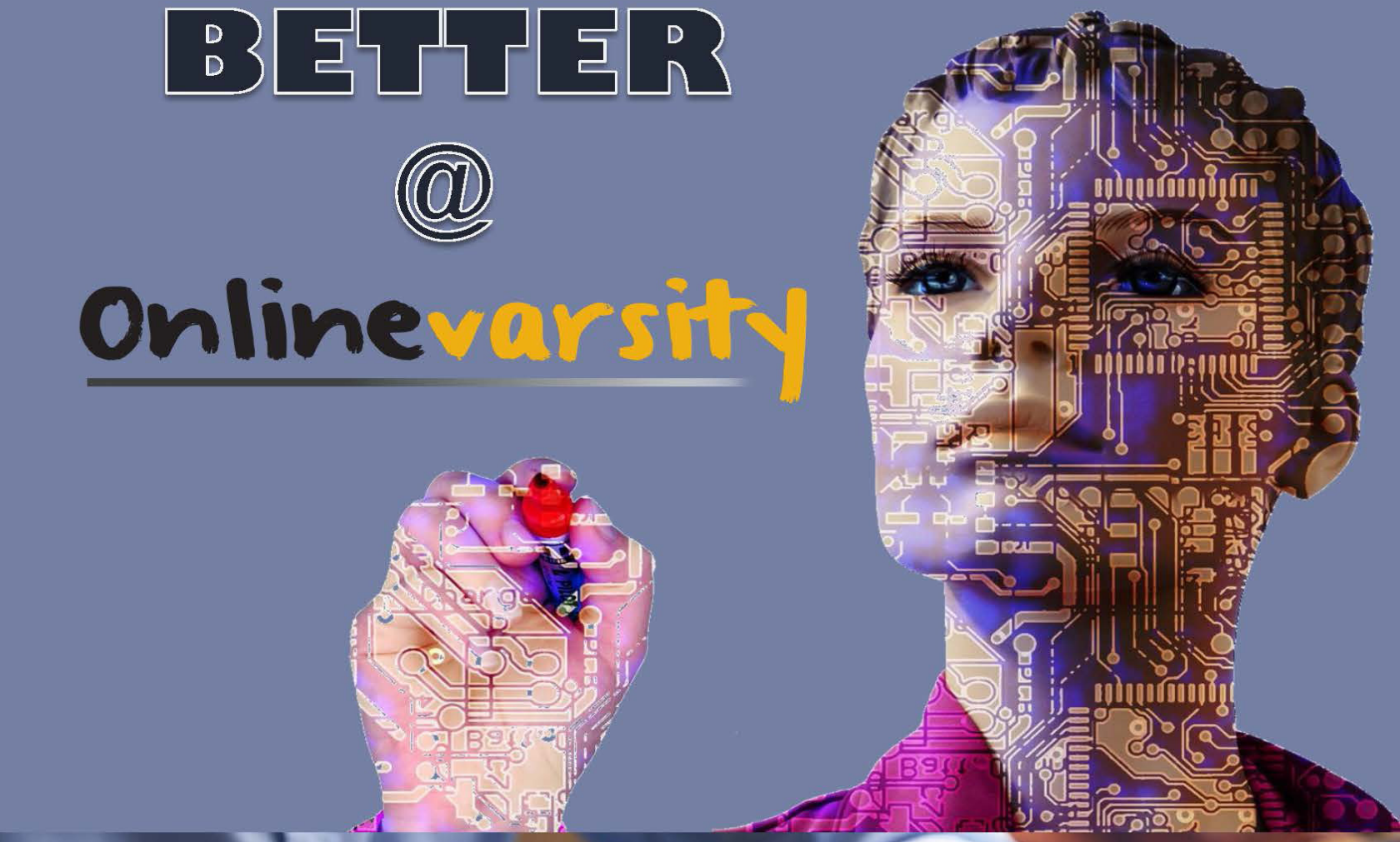
**Introduction** provides information related to the software industry. Process flow, suggested modules of the application and their description are also dealt with in the introduction.

**Documentation** describes different formats for collecting and maintaining the information. These can be used as guidelines.

**Case Study** provides source of information that can be used by the students while developing the project.

**Sample eProject** is for understanding the different aspects covered in project and the way of presenting them.

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# Introduction to the Book

The best way to learn something is to apply its principles and then to test them. Similarly, the best way to evaluate the knowledge you gained is to test its application through project work. This eProject Guide, which has been prepared following the best practices in the industry, will help you to have the experience of going through a live project and will teach you the essentials of successful development of IT projects.

The eProject Guide will help you to:

- Analyze a project
- Design the specifications of the project
- Develop the solution
- Maintain disciplined documentation for the work done
- Work in groups

The ability to work in a group is a very vital quality for anybody desiring to join the software industry. Your project group will consist of 3-4 members. The Faculty will assign you to a project group and select the Leader for the Group (Group Leader).

This eProject Guide reiterates the commitment of eProjects Team in keeping up its tradition of providing innovative, career oriented professional education.

Religiously following the given systematic approaches in this book would prepare the students to get the real life experience of handling projects, as the practices listed here have been extracted from the current industry norms. Thus, such an exercise would prepare you for joining the software development industry.

Wishing you the very best.

**eProjects Team**




# MANY COURSES ONE PLATFORM

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## for Android devices

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## 1. Introduction

### 1.1 Team

Implementation of any new computerized system involves a team of people. An ideal team consists of following members:

- **Project Manager**: The Project Manager coordinates the project. Apart from being knowledgeable about software, the person chosen as the Project Manager must have good writing skills. A Project Manager must have enough experience in this field so that he/she can be successful in managing the implementation. He/she is involved not only in team management, but also in activities such as resource allocation, project planning, reporting, and so forth; all of which form a part of the responsibilities.

- **Project Leader**: The Project Leader leads the project team. A Project Leader essentially decides which tasks are to be performed by each team member and how much time should be allotted to each project development phase.

- **Analyst**: The analyst studies the requirements of the system and defines the problem. The analyst determines the inputs, outputs, and processes involved in transforming those inputs into outputs. The analyst acts as the 'Technical Expert' and studies the technologies that are to be implemented to develop the project.

- **Developer/Programmer**: The developer builds the user interface according to the specifications prepared by the analyst. Next, the developer builds a prototype of the system. After receiving client approval on the prototype, the developer adds the necessary code to make the prototype a full-fledged system.

- **Tester**: The tester tests the functionality of the application. Test data is used to check if the program is able to execute without causing any errors. Test data may be live data extracted from existing records in the system or dummy data. The tester also verifies the integrated application's functionality with test data.

- **Implementation Engineer**: The Implementation Engineer ports the final product to the Client's computers. The Implementation Engineer will ensure that the installation process has been carried out accurately, and hands over the system to the client.

- **Maintenance Engineer**: The maintenance engineer is responsible for taking care of or maintaining the system that has been built. Maintenance includes extending troubleshooting support and performing software upgrades in case of changes in the external system.

## 1.2 Tips for a Good Application

The seven basic steps for creating an effective application are as follows:

1. Define your target audience
2. Organize your concepts, information, and material
3. Create a directory structure
4. Implement storyboarding – prepare a sketch of contents, sequence, and layout of the forms you intend to create
5. Build a prototype of your application
6. Test the prototype and implement the required changes
7. Deploy the application on the server, if applicable

Before designing an application, you must first conceptualize the application. The goals and the objectives must be clearly defined. Think about what the application goals and missions may be to inform, promote, educate, research and report, or to simply entertain. These goals must be defined and should be in line with the company's values and mission.

Annexure A shows a case study of a project.

## 1.3 Project Objectives

1. Develop a desktop application based on the problem specifications given
2. Design a professional-looking Graphical User Interface (GUI) for the application
3. Integrate all the modules to form a complete solution

## 1.4 Project Deliverables

Following are the deliverables that have to be submitted on the completion of the project:

1. Complete Application
2. User Manual
3. Installation Guide

## 1.5 Project Conduct

During project development, three meetings would be held within the team.

➢ **First Meeting**



In this meeting, students need to discuss the eProject Specification. They need to work upon:

- Problem Statement
- Table/File Structure
- Program/Code Specifications
- Reports to be generated

➢ **Review of First Meeting**

After the first meeting, students will review and discuss the details of the meeting. They will check if the structure is correct and the forms are designed as per the design specifications.

- **Second Meeting**



In this meeting, students will discuss the design. They need to consider the following:

- Understanding of the problem
- Design of the forms
- Validations required

- **Review of Second Meeting**

After the second meeting, students will review and discuss the details of the meeting. They will check whether the design of forms are correct and proper verification and validation rules are being implemented.

- **Third Meeting**

In this meeting, students will discuss and present the final application. They will consider the following:

- Implementing all the specifications mentioned in the previous meetings
- Form Design
- All the required validations

- Integrating all the project modules such as the forms, reports, and so forth
- User Manual and Installation Guide



## 2. Steps for a Great Project

- Analyze and understand the user requirements
- Do the detailed analysis for the project
- Identify the resources available
- Plan and maintain a schedule for the various activities to be done in the project
- Check and test all modules
- Document each and every detail

## Documentation Section

**In this section, we will look into various formats used for documenting the project.**

**This section contains different formats, which can also be referred to, in Annexure A, in the case study.**

**Students can use their own additional formats in the project documentation.**

## 3. Documentation Section

This format can be used as the first page of your project report, duly signed by the Center Academics Head.

This is to certify that

**Mr./ Ms** _____

Has successfully Designed & Developed

_____

Submitted by:

_____

Date of Issue:_____

Authorized Signature:_____

## 3.1 Problem Definition

**Problem Definition:**

In this form, you can define the problem statement. You can design a pointwise list of the requirements as mentioned in the eProject Specification.

Form No. 1/eProjects/PS/Ver 1.0

## 3.2 Customer Requirement Specification (CRS)

**Client:**

**Business/Project Objective:**

(can address organization/business overview, products, concerns, and expectations from the system)

**Inputs provided by the Client:**

- Inputs to the System
- Outputs from the System
- Process Involved in the System
- Expected Delivery Dates
- List of Deliverables

**Hardware Requirements:**

**Software Requirements:**

**Scope of the Work (in brief):**

Form No. 2/eProjects/CRS/Ver 1.0/1 of 1

## 3.3 Architecture and Design of the Project

You can provide the Architecture and Design for your eProject in this format.
You can refer to Annexure A Case Study to understand the use of Architecture and Design of the Project.

Form No. 3/eProjects/Design/Ver 1.0/1 of 4

## 3.4 Flow Chart

**A flow chart is a graphic representation of an algorithm or a process.**

You can provide the flow charts for your eProject in this format.
You can refer to Annexure A Case Study to understand the use of flow charts.

Form No. 3/eProjects/Design/Ver 1.0/2 of 4

## 3.5 Data Flow Diagram

**A Data Flow Diagram (DFD) is a graphical representation of the flow of data.**

You can provide the Data Flow Diagrams for your eProject in this format.
You can refer to Annexure A Case Study to understand the use of Data Flow Diagrams.

Form No. 3/eProjects/Design/Ver 1.0/3 of 4

## 3.6 Entity Relationship (ER) Diagram

**Entity Relationship is used to describe information needs and/or the type of information that is to be stored in the database.**

You can provide the Entity Relationship Diagrams for your eProject in this format.

Form No. 3/eProjects/Design/Ver 1.0/4 of 4

## 3.7 Database Design/Structure

**Table is a set of data elements (cells) that is organized, defined, and stored.**

Using this format, you can provide the Table Structure for your eProject in this format.
You can refer to Annexure A Case Study to understand the use of Table Structures.

Form No. 4/eProjects/DD/Ver 1.0/1 of 1

## 3.8 Task Sheet

| Project Ref. No.: | | Project Title: | Activity Plan Prepared By: | Date of Preparation of Activity Plan: | | | |
|---|---|---|---|---|---|---|---|
| Sr.No. | Task | | | Actual Start Date | Actual Days | Team Mate Names | Status |
| | | | | | | | |

Form No. 5/eProjects/TS/Ver 1.0/1 of 1

## 3.9 Checklist of Validations

| Option | Validated |
|---|---|
| Do all numeric variables have a default value of zero? | |
| Does the administrator have all the rights to create and delete the records? | |
| Are all the records properly fed into the appropriate database? | |
| Have all the modules been properly integrated and are completely functional? | |
| Have all the Design and Coding Standards been followed and implemented? | |
| Is the GUI design consistent all over? | |
| Is the navigation sequence correct through all the forms/ screens in the application? | |
| Is exception handling mechanism implemented in all the screens? | |
| Are all the program codes working? | |

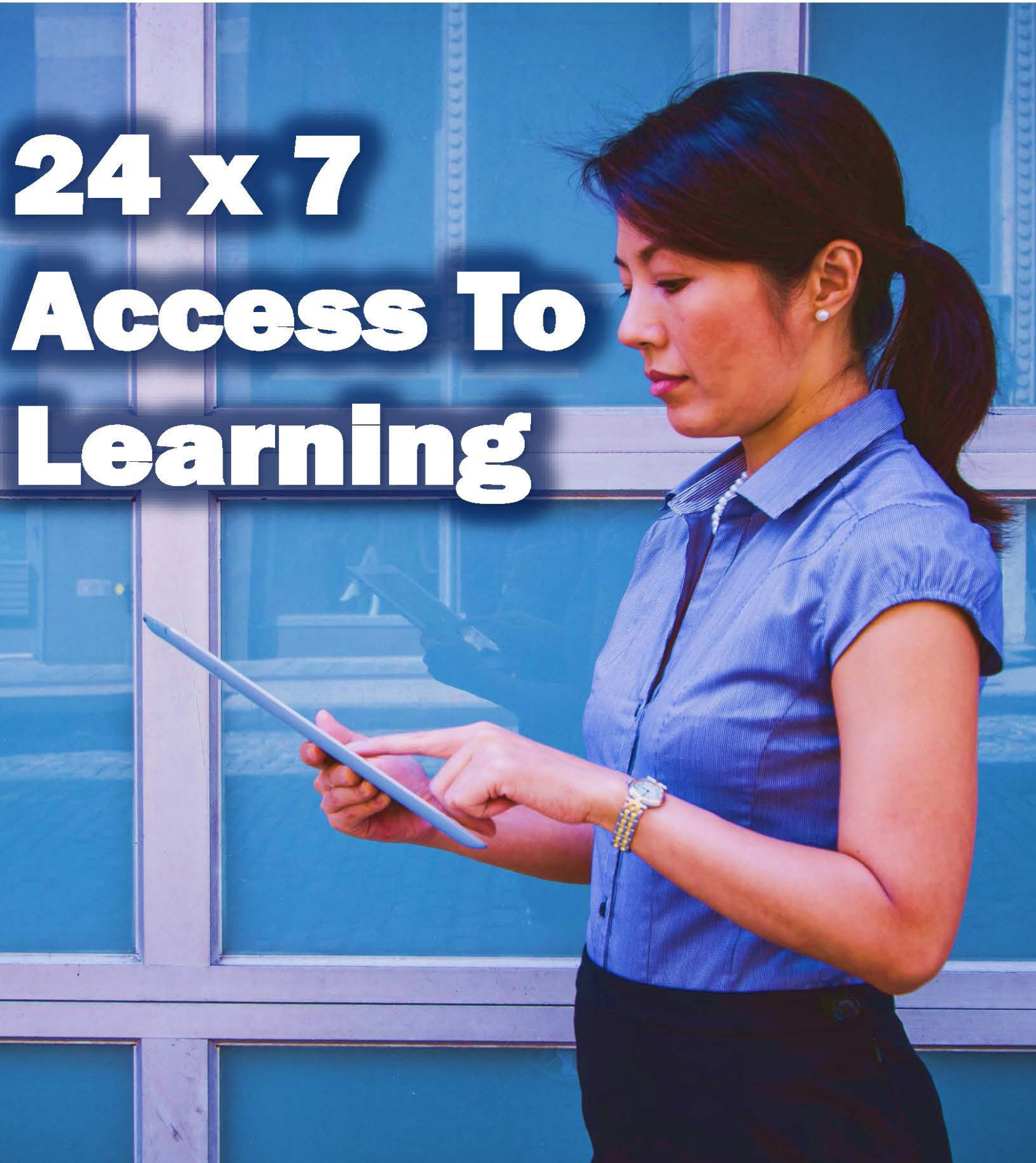
Form No. 6/eProjects/VAL/Ver 1.0/1 of 1

## 3.10 Submission Checklist

| Sr. No. | Particulars | Yes | No | NA | Comments |
|---|---|---|---|---|---|
| 1. | Are all the users able to search for a particular record? | Yes | | | |
| 2. | Are all the old records properly saved and retrieved when required? | Yes | | | |
| 3. | Have all the modules been properly integrated and are completely functional? | Yes | | | |
| 4. | Are the GUI contents devoid of spelling mistakes? | Yes | | | |
| 5. | Is the application user-friendly? | Yes | | | |
| 6. | Is the project published properly into a setup file? | Yes | | | |

Form No. 7/eProjects/CKL/Ver 1.0/1 of 1

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# Annexure A: CASE STUDY

**Shoppaholics Pvt. Ltd**. is a popular fashion clothing chain of stores in Wisconsin. The store sells all popular brands of clothes and accessories. The store also offers discounts and organizes sale at the end of every year. With the increasing demand and customer base, the management of the store had created a Website last year from where the customers can get information about the latest updates of sale and new stock related details of their favorite brands. Currently, the customers who wish to view or purchase anything from the site have to either call or visit the outlet to place an order or place their order online on the Website **Shoppaholics.com**. However, the management has observed lately that customers have been facing difficulties in using the Website especially on mobile devices. To provide the flexibility to the customer and to improve order management, the company has decided to develop a mobile application. The management conducted a survey on popular mobile devices and found that majority people use Android based phones. Hence, they have decided to create and launch an Android application.

The customers would download the application on their Android phone and create an account. After that, the customer will be able to perform all the functions of viewing and ordering the products similar to their Website.

## Proposed Solution:

The mobile application can be created based on the following specifications:

- The project will be Shoppaholics Mobile Application created using Android 6.0, Android Studio 2.1, and SQL Lite.
- SQLite will be used as the database for storing all the required information. Appropriate tables will be created to store information related to products, customers, orders, and employees.
- The application will be used by customers as well as employees of Shoppaholics.
- The users will need to create an account to view the products and place orders.
- The users will be sent an SMS on placing an order and making payment.

The application will have three categories of users accessing it:

- Administrator
- Employees of Shoppaholics
- Customers of Shoppaholics

**Administrator**:

The administrator should be able to perform the following operations:

- Create new employee logins
- View product details
- View reports
- View customer feedback

**Employees**:

The employees should be able to perform the following operations:

- Login to the application with company provided login ids
- View and update the order details
- Send sms to registered customers

**Customer**:

Using the application, they can perform the following tasks:

- Register with Shoppaholics
- View the products
- Place an order
- Fill feedback form

## Project Specifications:

The application is to be developed for Shoppaholics company using Android 6.0 and SQLite.

The application should perform the following functions:

**Administrator section**:

- Add/edit/delete product details
- Add/edit/delete/activate/deactivate customers
- Add/edit/delete/activate/deactivate employees
- Add/edit/delete/activate/deactivate administrator
- Generate a daily report about products and orders

**Employee section**:

- View the orders
- Confirm orders and send sms

**Customer Section**:

- Register with Shoppaholics
- View the categories of products
- Place order for a product
- Fill feedback form

**Reports**:

- Product inventory
- List of orders placed
- Sales report
- Employee-wise product sales details

**Hints for developing the Solution:**

1. User Interface for Application: Use interface builder to design the user interface using View, Layouts, and the various UI components.

    - **Customer Registration Form**: Design the Customer Registration form, which includes Customer name, address, mobile, e-mail, username, password, and so on.

    - **Product Display Form**: Design the Product Display form which displayed the categories of products. Clicking a particular category should display another form listing all products available in that category. When user selects a particular product, another screen should open showing all details of that product such as name, price, description, and so on.

    - **Customer Order Form**: Design the Customer Order form which is opened when user clicks Order button on a particular product. This form should accept details such as quantity, size, and so on based on the type of product selected by the user.

    - **Report**: Design the Report screen, which displays a drop-down to view a particular type of report. On selecting the report type, relevant input should be required. For example, if the Sales Report option is selected, it should ask for the date, month, or year for which sales report is required.

2. **SMS feature**: Customer should be sent an SMS after customer registration and confirmation of the order request. Also, they should receive the promotional and sale messages about products through SMS. For this, the developer can use SMSManager.

3. **Database**: Create the database using SQLite. Database can also be created using other DBMS available.

## ACKNOWLEDGEMENT

I would like to acknowledge all those who have given moral support and helped me make the project a success.

I wish to express my gratitude to the eProjects Team at the Head Office, who guided and helped me. I would also like to express my gratitude to all the staff members of my center for not only providing me with the opportunity to work with them on this project, but also for their support and encouragement throughout the process.

I also express my sincere gratitude to the my project guide at the organization, for his/her valuable guidance and support for the completion of this project.

And finally, I would like to offer many thanks to all my colleagues for their valuable suggestions and constructive feedback.

**Problem Definition**

After reading the project specifications, the developer states the scope of the project very briefly.

The **Problem definition** in the Shoppaholics Mobile Application would be: With the rise in the customer base, the company is finding it difficult to handle so many requests on the Website. Hence, there is a need to offer software solutions to offset cost by substituting paper based expenses with increased control, efficiency, productivity, and live documentation.

This Software will enable user to view and order products as well as maintain the customer data.

Also, the application will be capable of storage, retrieval, and manipulation of data in appropriate database files.

Form No. 1/eProjects/PS/Ver 1.0

**Customer Requirement Specification (CRS)**

**Client: Shoppaholics Mobile Application**

**Business/Project Objective:**

(can address organization/business overview, products, concerns, and expectation from the system)

- The application will be used by customers as well as employees of Shoppaholics.
- The users will need to create an account to view the products and place orders.
- The users will be sent an SMS on placing an order and making payment.

**Inputs provided by the client:**

- Inputs to the existing system
- Outputs from the existing system
- Process involved in the application
- Expected delivery dates
- List of deliverables

**Hardware Requirements:**

- Intel Core i3 Processor or higher
- 4 GB RAM or above
- Color SVGA
- 120 GB Hard Disk space

**Software Requirements:**

- Microsoft Windows 7 or Microsoft Windows 8.1 or higher
- Android 6.0, Android Studio 2.1, and SQL Lite

Form No. 2/eProjects/CRS/Ver 1.0/1 of 2

**Scope of the Work (in brief):**

Depending on the decision taken by the management regarding the Shoppaholics Mobile Application, following are some of the requirements based on which the software needs to be developed:

- Login
- Customer Registration
- Order Products

Form No. 2/eProjects/CRS/Ver 1.0/2 of 2

### Architecture and Design of the System

Figure 1.1 displays the application architecture.

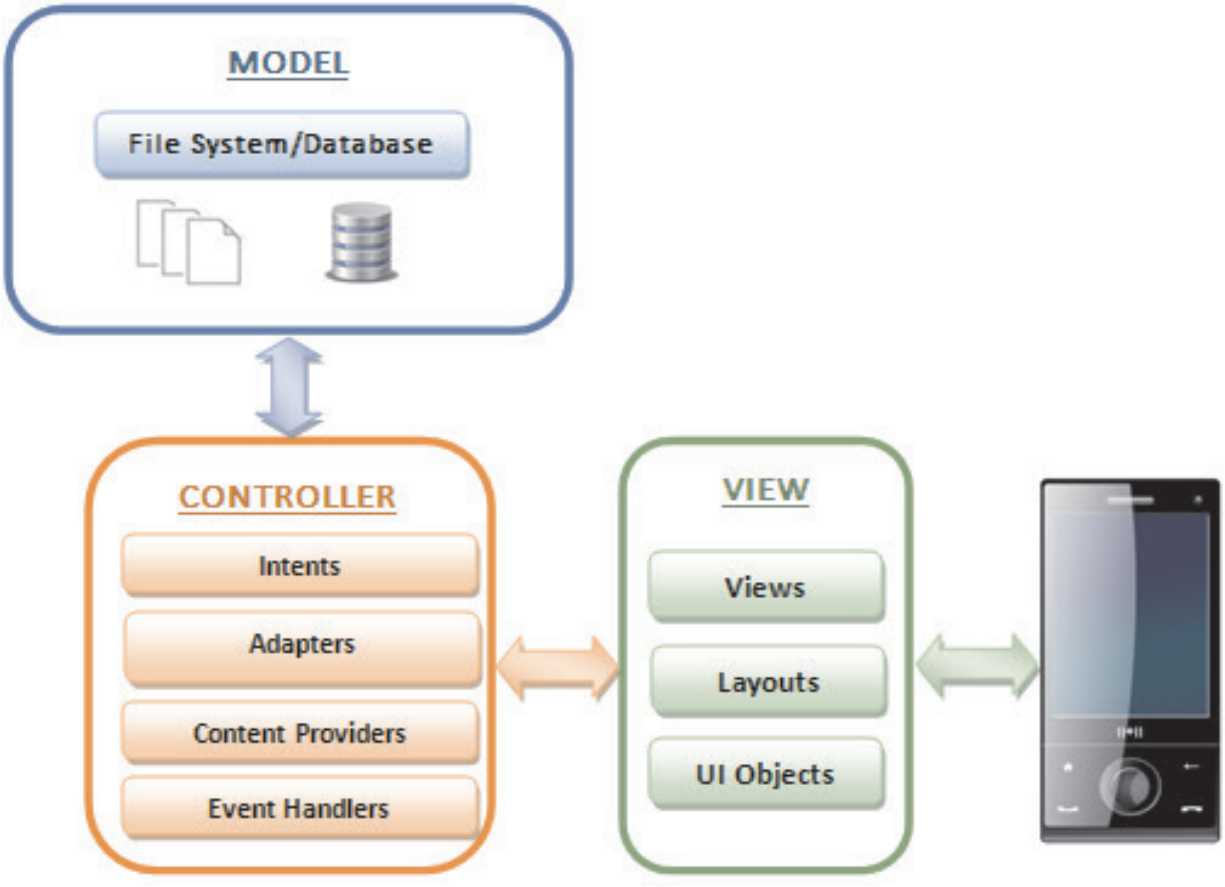



**Figure 1.1: Application Architecture**

Form No. 3/eProjects/Design/Ver 1.0/1 of 6

## Data Flow Diagram

Figure 1.2 shows Level 0 DFD.

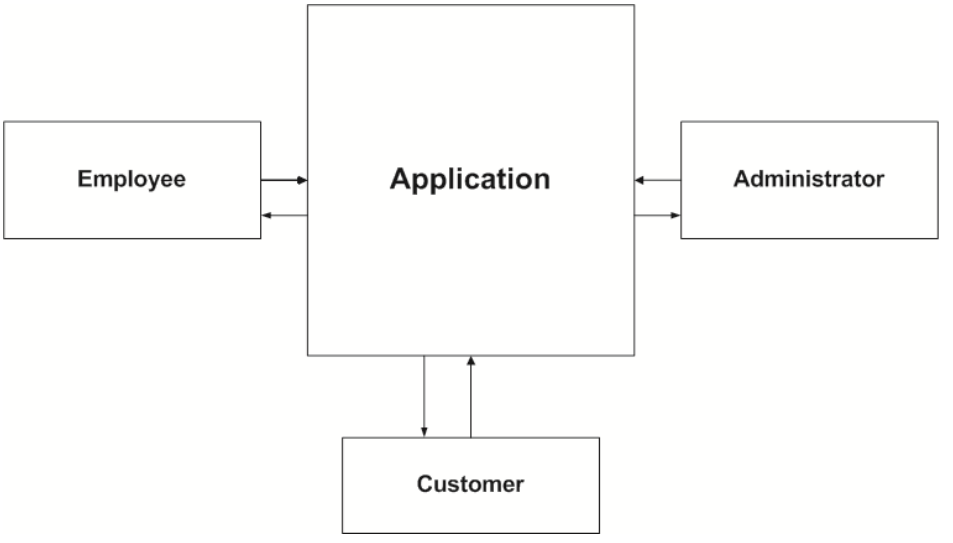



**Figure 1.2: Level 0 DFD**

Figure 1.3 displays the first level of Data Flow Diagram.

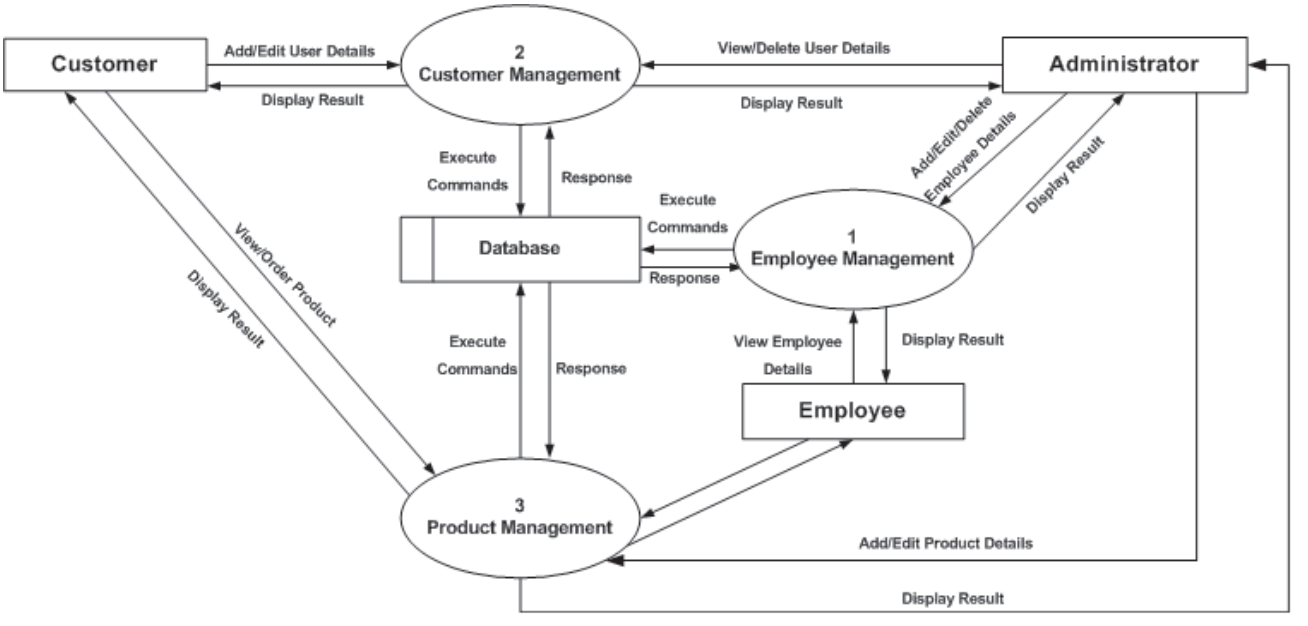



**Figure 1.3: Level 1 DFD**

Form No. 3/eProjects/Design/Ver 1.0/2 of 6

Figure 1.4 shows Level 2 DFD of Employee Management module.

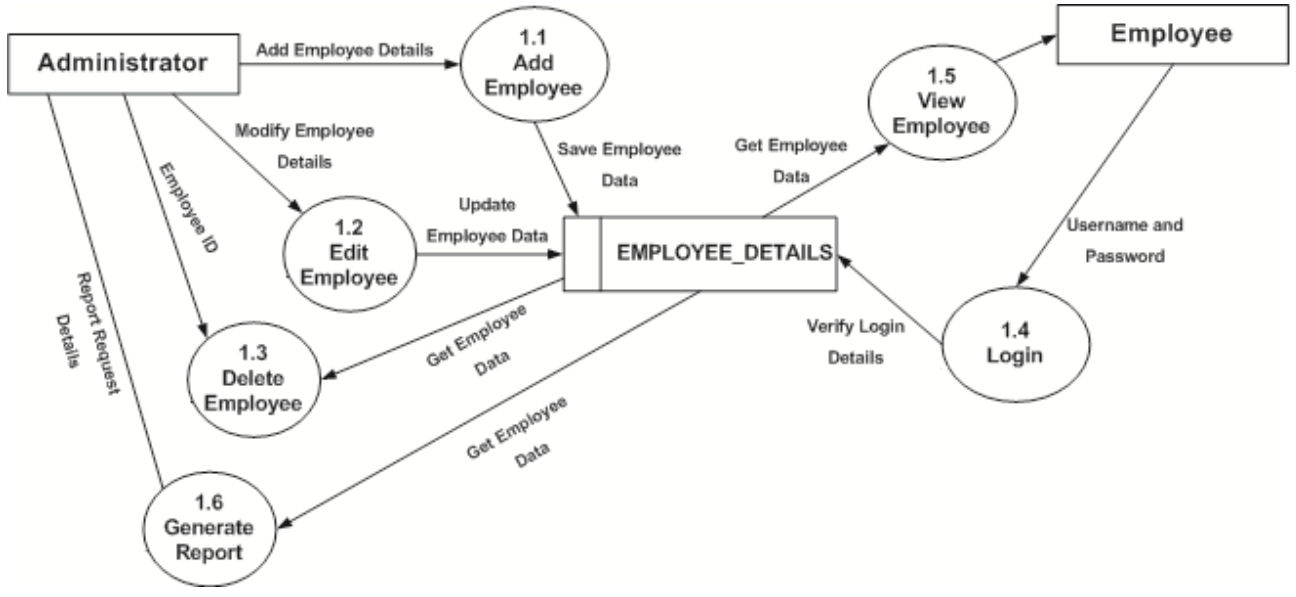



**Figure 1.4: Level 2 DFD - Employee Management**

Figure 1.5 shows Level 2 DFD of Customer Management module.

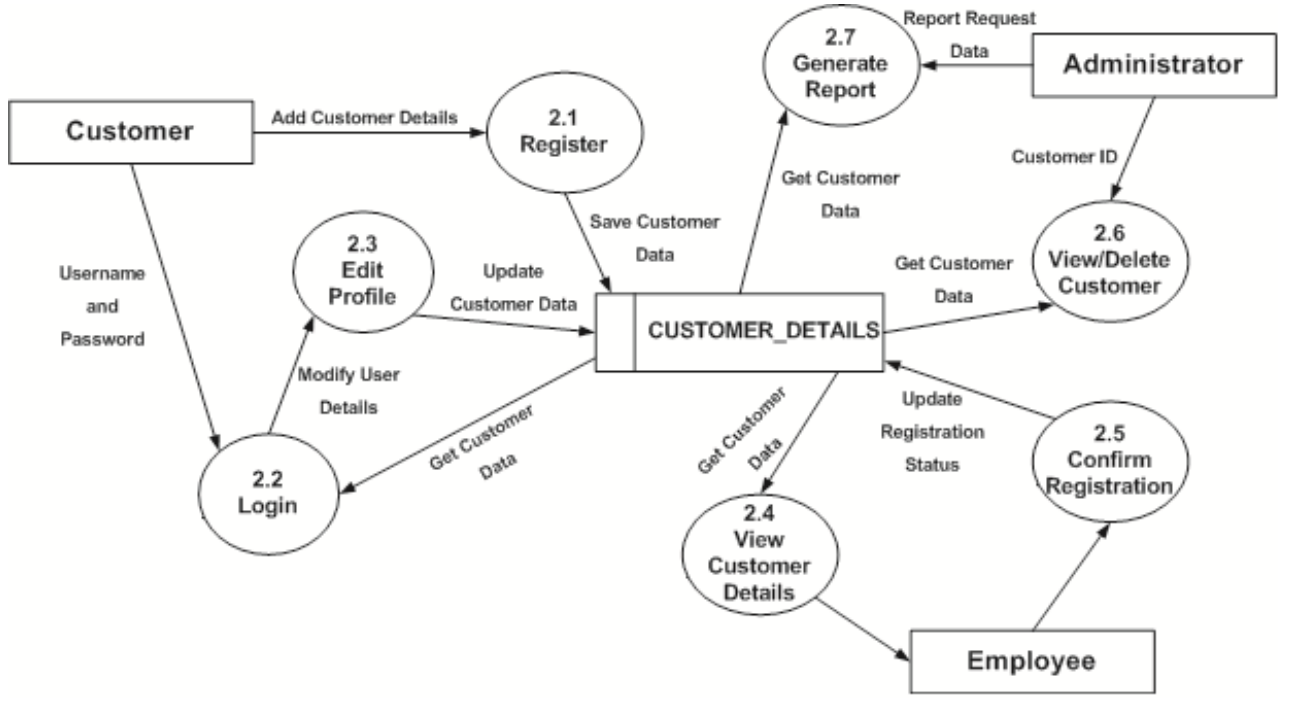



**Figure 1.5: Level 2 DFD – Customer Management**

Form No. 3/eProjects/Design/Ver 1.0/3 of 6

Figure 1.6 shows Level 2 DFD of Product Management module.

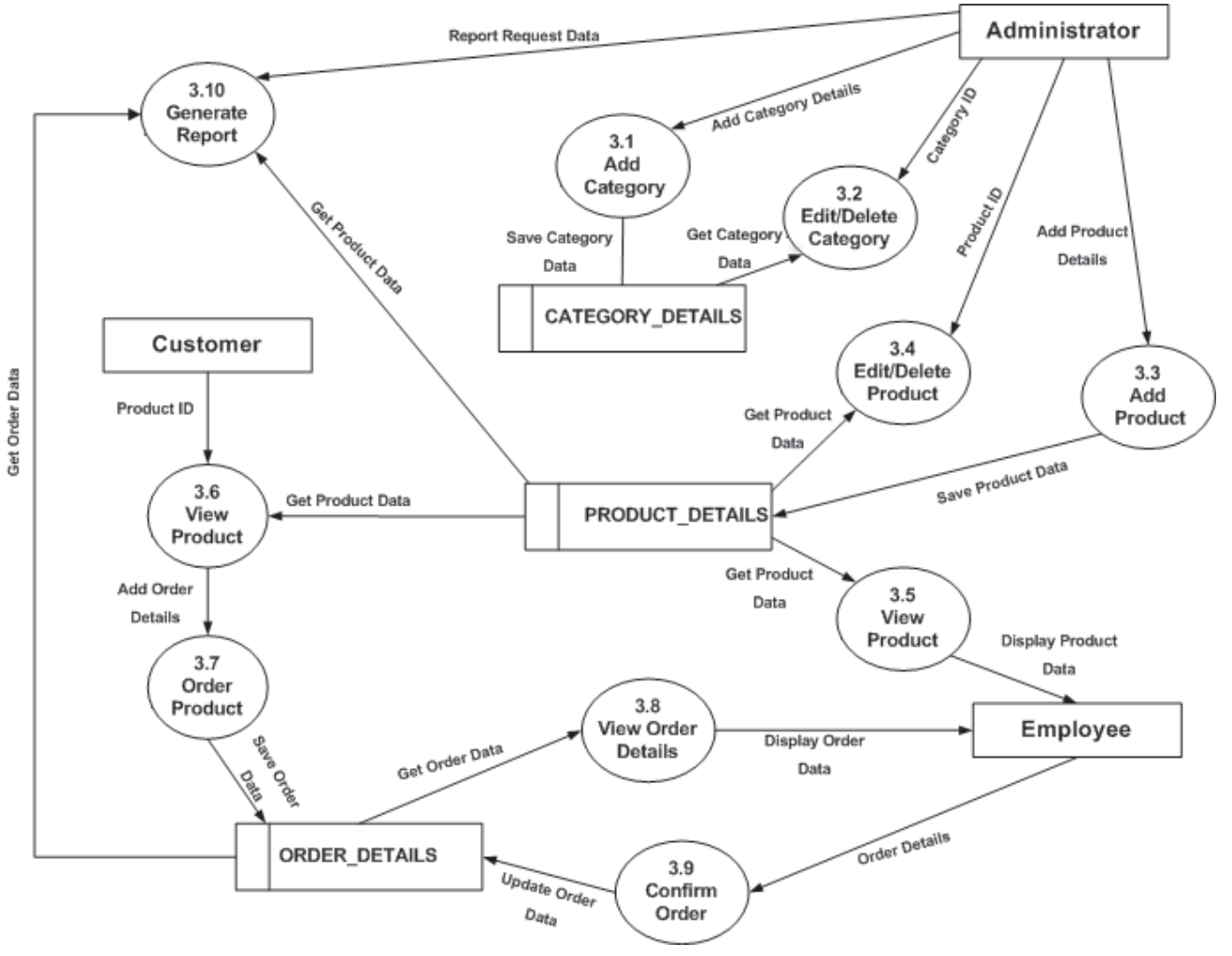



**Figure 1.6: Level 2 DFD – Product Management**

Form No. 3/eProjects/Design/Ver 1.0/4 of 6

Figure 1.7 shows the application flow.

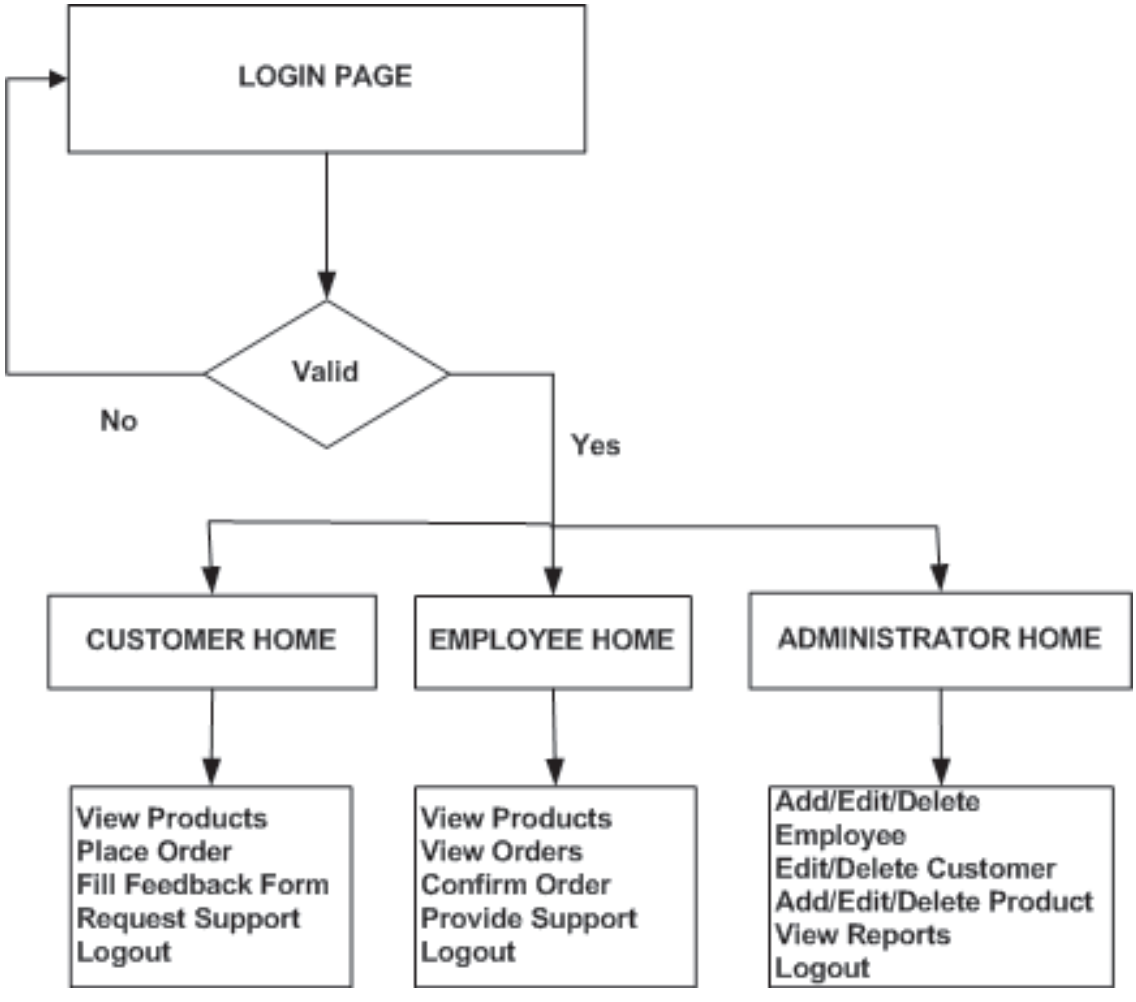



**Figure 1.7: Application Flow**

Figure 1.8 shows the database design and table relationships.

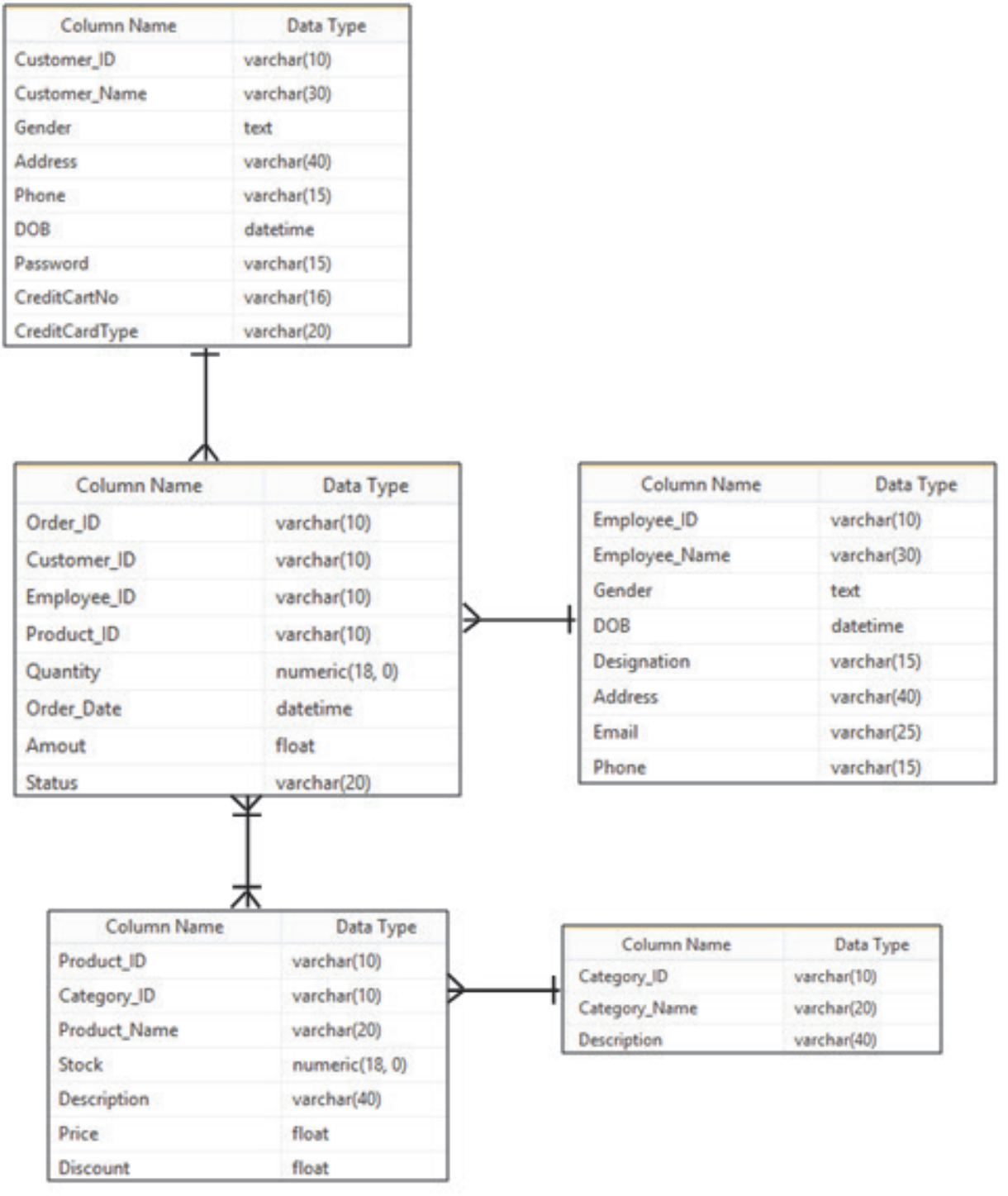



**Figure 1.8: Database Design and Table Relationships**

Form No. 3/eProjects/Design/Ver 1.0/6 of 6

Database structure of the different tables is given in the following sections:

- **Employee_Details**

  The Employee_Details table stores the employee details.

| Field Name | Data Type | NULL | PK/FK | Reference Table | Description |
|---|---|---|---|---|---|
| Employee_ID | VARCHAR(10) | NO | PK | | Stores employee's ID |
| Employee_Name | VARCHAR(30) | NO | | | Stores employee's name |
| Gender | TEXT | NO | | | Stores the gender of the employee |
| DOB | DATETIME | NO | | | Stores the birth date of the employee |
| Designation | VARCHAR(15) | NO | | | Stores employee's designation |
| Address | VARCHAR(40) | NO | | | Stores employee's address |
| Email | VARCHAR(20) | NO | | | Stores employee's email |
| Phone | VARCHAR(15) | NO | | | Stores employee's phone number |

- **Customer_Details**

The Customer_Details table stores the customer details.

| Field Name | Data Type | NULL | PK/FK | Reference Table | Description |
|---|---|---|---|---|---|
| Customer_ID | VARCHAR(10) | NO | PK | | Stores customer's ID |
| Customer_Name | VARCHAR(30) | NO | | | Stores customer's name |
| Gender | TEXT | NO | | | Stores customer's gender |
| Address | VARCHAR(40) | NO | | | Stores customer's address |
| Phone | VARCHAR(15) | NO | | | Stores customer's phone number |
| DOB | DATETIME | NO | | | Stores customer's date of birth |
| Password | VARCHAR(15) | NO | | | Stores customer's password |
| CreditCardNo | VARCHAR(16) | NO | | | Stores customer's credit card number |
| CreditCardType | VARCHAR(20) | NO | | | Stores the credit card type |

- **Product_Details**

The Products_Details table stores product details.

| Field Name | Data_type | NULL | PK/FK | Reference Table | Description |
|---|---|---|---|---|---|
| Product_ID | VARCHAR(10) | NO | PK | | Stores product ID |
| Category_ID | VARCHAR(10) | NO | FK | Category_Details | Stores product category |
| Product_Name | VARCHAR(20) | NO | | | Stores product name |
| Stock | NUMBER | NO | | | Stores the product's available quantity |
| Description | VARCHAR(40) | NO | | | Stores the product description |
| Price | FLOAT | NO | | | Stores the product price |
| Discount | FLOAT | NO | | | Stores the discount available on the product |

- **Category_Details**

The Category_Details table stores the list of product categories.

| Field Name | Data_type | NULL | PK/FK | Reference Table | Description |
|---|---|---|---|---|---|
| Catetory_ID | VARCHAR(10) | NO | PK | | Stores the category ID |
| Category_Name | VARCHAR(20) | NO | | | Stores category name |
| Description | VARCHAR(40) | NO | | | Stores category description |

- **Order_Details**

The Order_Details table stores the transaction details of the orders.

| Field Name | Data_type | NULL | PK/FK | Reference Table | Description |
|---|---|---|---|---|---|
| Order_ID | VARCHAR(10) | NO | PK | | Stores the order ID |
| Customer_ID | VARCHAR(10) | NO | FK | Customer_Details | Stores the customer ID |
| Employee_ID | VARCHAR(10) | NO | FK | Employee_Details | Stores the employee's ID to whom order has been assigned |
| Product_ID | VARCHAR(10) | NO | FK | Product_Details | Stores the product ID |
| Quantity | NUMBER | NO | | | Stores the quantity |
| Order_Date | DATETIME | NO | | | Stores the date and time of order placement |
| Amount | FLOAT | NO | | | Stores the payable amount |
| Status | VARCHAR(20) | NO | | | Stores the status of the order |

**Task Sheet**

| Project Ref. No.: Shoppaholics Mobile Application | | Project Title: | Activity Plan Prepared By: | Date of Preparation of Activity Plan: | | | |
|---|---|---|---|---|---|---|---|
| Sr.No. | Task | | | Actual Start Date | Actual Days | Team Mate Names | Status |
| | | | | | | | |

Form No. 5/eProjects/TS/Ver 1.0/1 of 1

**Checklist of Validations**

| Option | Validated |
|---|---|
| Do all numeric variables have a default value of zero? | Yes |
| Does the administrator have all the rights to create and delete the records? | Yes |
| Are all the records properly fed into the appropriate database? | Yes |
| Have all the modules been properly integrated and are completely functional? | Yes |
| Have all the design and coding standards been followed and implemented? | Yes |
| Is the design consistent all over? | Yes |
| Is the navigation sequence correct through all the screens in the application? | Yes |
| Is exception handling mechanism implemented in all the screens? | Yes |
| Are all the program codes working? | Yes |

Form No. 6/eProjects/VAL/Ver 1.0/1 of 1

**Submission Checklist**

| Sr. No. | Particulars | Yes | No | NA | Comments |
|---|---|---|---|---|---|
| 1. | Have all the modules been properly integrated and are they completely functional? | Yes | | | |
| 2. | Does each unit meet its objective and purpose?<br><br>Are all the validations happening as specified in Process Design? | Yes | | | |
| 3. | Have all Design and Coding standards been followed and implemented? | Yes | | | |
| 4. | Is the GUI design consistent all over? | Yes | | | |
| 5. | Are the codes working as per the specification? | Yes | | | All the modules are properly tested. |
| 6. | Does the application's functionality resolve the client problem, and satisfy his/her needs completely? | Yes | | | |
| 7. | Have the hardware and software been correctly chosen? | Yes | | | |
| 8. | Additional features and utilities that give value addition to the entire project. | Yes | | | |

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# 4. User Guide

A. **System Requirements:**

| No. | Items | Description |
|---|---|---|
| 1 | Operating System | Microsoft Windows 7 or Microsoft Windows 8.1 or higher |
| 2 | Database | SQL Server 2012/2016 |
| 3 | Software | Android 6.0, Android Studio 2.1, and SQL Lite |

B. **Install and Run Application:**

**Step 1:** Install SQL Server 2012/2016 on the local system.

**Step 2:** Create appropriate database and tables as per the specifications given.

**Step 3:** Execute the application.